

Verilog literals

Constants show up everywhere in RTL, reset values, parameter defaults, and compare masks

How literals are built

- A sized literal has three parts
- Eight apostrophe h F F is eight bits of all ones
- An unsized literal like apostrophe b one zero one zero still has a base but lets the tool
- Add s after the apostrophe for signed interpretation
- Underscores between nibbles are cosmetic; they do not change the value

Browser lab

igital Design and Verification Platform

HomeToolsVerilog literals

Challenges

0 / 28 cleared

4-bit binary

Intro · Enter a literal whose bits are exactly 1010 (width 4).

Hex FF

Intro · Decode 8'hFF — unsigned should be 255.

Sized decimal 'd

Intro · Enter 16'd42 — unsigned should be 42, width 16.

Plain decimal 42

Intro · Enter plain unsigned decimal 42 (no quote).

Signed -1

Intro · Enter 8'sb1111_1111 or 8'sd-1. Signed value should be -1.

Uppercase base

Core · Enter 8'H2A (capital H). Same bits as 8'h2A.

Underscore nibble

Core · Enter 32'hDEAD_BEEF (underscores allowed).

Truncation warn

Core · Enter 4'hFF so the tool truncates to 4 bits and warns.

Zero extend

Core · Enter 8'b1010 — should zero-extend to 0000_1010.

Octal

Core · Enter 6'o17 (octal). Bits should be 001_111.

Unsigned binary

Core · Enter 'b1010 (unsigned). Bits should be 1010.

Unsigned octal

Core · Enter 'o17 (unsigned). Bits should be 001_111.

4-bit binary: Enter a literal whose bits are exactly 1010 (width 4).

Start selected

Show hint

Next

Stop

Reset progress

Idle

Literal

VERILOG / SYSTEMVERILOG NUMBER

8'h2A

Decode

4'b1010

8'hFF

8'h2A

8'H2a

16'd42

8'sd-1

8'sb1111_1111

4'shF

6'o17

12'hABC

'b1010

'o17

'd255

'hF

42

5'b10x0z

4'b10??

32'hDEAD_BEEF

Literal OK — bit pattern shown below.

Starter example loaded.

Anatomy

Size

8

Signed

no

Base

'h (16)

Digits

undefined

0

0

1

0

1

0

1

0

MSB (left) outlined — sign bit when interpreting as signed.

Decoded values

bits

0010_1010

width

8

base

'h

unsigned

42

signed

42

hex

2A

Copy bits

Print

Clear saved

Load starter example

Open radix converter

Verilog literals lab starter

Real Verilog practice

```
wsl - module02-verilog-literals/examples/navigation

# Track A - real Unix
# real Verilog session (Track A)

$ cat examples/literals/literals.v
// literals.v - sized bases for Track A practice
module literal_demo;
    localparam [7:0] MASK_FF = 8'hFF;
    localparam [3:0] PATTERN = 4'b1010;
    localparam [15:0] COUNT   = 16'd42;
    localparam [7:0] NEG1     = 8'shFF;
endmodule

$ iverilog -t null examples/literals/literals.v
(syntax check passed)
```

Pitfalls to watch

- Do not ignore width, four apostrophe h F F truncates to four bits
- Do not treat signed and unsigned as interchangeable
- X and Z are for simulation and unknown states
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and finish a few challenges
- On real Verilog, write or edit a minimal sketch with at least two different bases
- When you are ready